

Arsh Tiwari

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EDUCATION

Bhagwan Parshuram Institute of Technology

Bachelor of Technology in Computer Science and Engineering; **CGPA: 9.15**

Delhi, India

July 2023 - June 2027

WORK EXPERIENCE

MathonGo

Software Developer Intern

July 2025 – Jan 2026

Bangalore, India

- Revamped the Quizrr platform (1L+ users) using **React.js**, improving page load time by **35%** and boosting user engagement through consistent, scalable UI architecture.
- Built the **Notebooks feature end-to-end**, designing reusable components and integrating backend APIs for data persistence, increasing feature adoption by **25%** and improving session retention.
- Implemented **request deduplication** for API calls by caching in-flight requests, reducing duplicate network calls by **40%** and improving performance of data-heavy workflows.
- Integrated **RabbitMQ** to handle asynchronous tasks of note-saving and analytics, offloading heavy operations from synchronous APIs and reducing API response time by **25%** during peak usage.

Eterna Infotech Pvt. Ltd.

Software Developer Intern

Nov 2024 – Mar 2025

Remote

- Developed and scaled an E-learning platform (10K+ users) using **React.js**, improving initial load time by **40%** via code splitting and modular architecture.
- Optimized bundle size by enabling compression using **Webpack**, reducing JavaScript payload by **30%**.
- Reduced MongoDB query latency by 35% by adding compound indexes on frequently queried fields (userId, courseId) used in progress tracking and dashboard APIs.
- Implemented CDN-based asset delivery for Australian users, reducing latency by **35%** and improving load performance.
- Implemented cache-aside pattern for frequently accessed course data using Redis, reducing repeated database hits by **30%** during high traffic.

ACHIEVEMENTS

- Grand Finalist at Smart India Hackathon 2025, ranked among the **top 5 teams** out of 68,000+ teams nationwide.
- Achieved **Pupil** rank on **Codeforces** (Max Rating: **1233**), ranking in the **top 25% of active users globally**.
- Ranked in the **top 6.25% globally** on **LeetCode** (Rating: **1842**), solving **350+ problems** with 78%+ acceptance.
- Earned **2 Star** rating on **CodeChef** (Max Rating: **1412**), placing in the **top 30%** of competitive programmers.
- Secured **94 percentile** in JEE Main 2023, ranking in the **top 4% among 2M+ candidates** nationwide.
- Winner of **MapmyIndia Hackathon** at IIT Delhi, among 720+ teams in a 24-hour national-level competition.

PROJECTS

GOSEARCH | Golang, Gin, REST API, React, Redis, PostgreSQL | [Link](#) | [Code](#)

- Achieved **1.8 ms average query latency** on a 52K-term dataset by building a full-text search engine in Go using **TF-IDF ranking and inverted indexing**, reducing query complexity from O(N) to O(k).
- Enabled **highly concurrent search throughput** by designing a thread-safe index using **sync.RWMutex**, supporting unlimited parallel reads with zero race conditions (validated via Go race detector).
- Improved scalability and reliability by introducing **Redis caching (1 ms repeated queries)**, PostgreSQL persistence, and rate-limiting middleware handling **200+ req/sec bursts** without failure.

APPLYSTACK | React, Node.js, Express.js, MongoDB, Redis, RabbitMQ, bcrypt, Docker | [Link](#) | [Code](#)

- Streamlined job application workflows by building a full-stack platform with **React, Node.js, and MongoDB**, reducing candidate management overhead and increasing application speed by **25%**.
- Containerized **4 microservices** using **Docker & Docker Compose** to ensure **100% environment parity**, effectively eliminating deployment configuration errors and local-to-production environment drift.
- Reduced database load by **60%** using **Redis caching**, ensured **high availability**, and improved API response times by **10%** by offloading weekly report generation and email delivery to **RabbitMQ** based asynchronous workers.

SKILLS SUMMARY

Programming Languages: C++, Golang, SQL, JavaScript, TypeScript, HTML5, CSS, Python

Frameworks and Databases: React.js, Next.js, Node.js, Express.js, MERN Stack, WebSockets, REST API, gRPC, RabbitMQ, Redis, MySQL, PostgreSQL, MongoDB

Coursework: Object-Oriented Programming, Database Management System, Operating System, Computer Networks

Tools & Platforms: Amazon Web Services (Foundational), Docker, Git & GitHub, Postman, Vercel, Webpack

POSITIONS OF RESPONSIBILITY

Technical Team Lead at Pears Global Hackathon 2025

Led technical team to build official website and managed 2k+ registrations from 12+ countries

Delhi, India

March 2025